**Assignment 4** ID5103: Mathematical Finance Due Sunday, Oct 11

In this assignment, we explore variable interest rates in financial markets. As discussed in class, variable interest rates are generally modelled as random processes driven by a random experiment such as a series of coin tosses. This gives rise to a binomial tree model for interest rates as well.

Variable interest rates can be specified in many different ways. One way is to specify the **short rate** $\mathbb{R}_n$ that applies for one time period between $n$ and $n+1$. The interest rate $\mathbb{R}_n$ is a random variable that depends on the first $n$ coin tosses and is determined at the end of $n$ time periods, and is applicable for the $(n+1)^{\text{th}}$ time period between $n$ and $n+1$.

Closely related to the short rate process $\mathbb{R}_n$ is the discount process $\mathbb{D}_n$ which is defined as

$$\mathbb{D}_n = \frac{1}{1+\mathbb{R}_0}\frac{1}{1+\mathbb{R}_1}\cdots\frac{1}{1+\mathbb{R}_{n-1}}\ . \tag{1}$$

This is basically the compounded discounting of an asset price at time $n$ to time 0. The discounting happens over the period $n-1$ to $n$, $n-2$ to $n-1$, ..., 2 to 1, and finally 1 to 0. The product of factors on the right hand side indicate this successive discounting from time $n$ to time 0 using the short rates that apply over each time interval.

Once we have specified the interest rate model, we can discuss bonds. Bonds are contracts which have a particular maturity time, say $M$, and fixed non-random payments at the maturity time and possibly at some intermediate times before maturity. The payment at maturity is called the face value of the bond and the payments at intermediate times are called coupons.

The simplest bonds are those which pay a fixed face value at maturity and have no coupons. These are called **zero coupon bonds**, ZCB for short, and are labelled by the maturity time $M$ and face value. By convention, we take the face value of a single ZCB to be 1. For bonds of face values other than 1, we can consider those many ZCBs with face value 1 (note that this number can be a fraction or an irrational number, and we allow such 'numbers' of bonds in an ideal mathematical world).

We denote zero coupon bonds with maturity time $M$ and face value 1 as $\text{ZCB}_M$. In general, we assume that bonds of all possible maturities exist $M = 1, \ldots, N$ where $N$ is some large time called the **horizon time**.

Unlike the case of derivatives based on stocks, bonds are assets which are not based on tradable assets like stocks, but are rather defined in terms of interest rates which are not tradable. Because of this, it is difficult to replicate the bond in terms of more fundamental primary assets which would then give rise to a risk-neutral probability measure (like we had for derivatives based on stocks). Thus, to obtain no-arbitrage prices of bonds, we start with

an arbitrarily chosen probability measure and brand this the 'risk-neutral measure'. We then **define** (as opposed to derive) the no-arbitrage bond price at times before maturity to be given in terms of conditional expectations using this *ad hoc* risk neutral measure.

Let $\mathbb{B}_{n,M}$ be the no-arbitrage price of a $\text{ZCB}_M$ at time $n \leq M$. Since the face value of $\text{ZCB}_M$ is 1, this means $\mathbb{B}_{M,M} = 1$. The no-arbitrage price is then **defined** by the formula

$$\mathbb{D}_n \mathbb{B}_{n,M} = \widetilde{\mathbb{E}}[\mathbb{D}_M \mathbb{B}_{M,M} \mid \mathcal{F}_n] = \widetilde{\mathbb{E}}[\mathbb{D}_M \mid \mathcal{F}_n] \quad \text{for} \quad 0 \leq n \leq M \ , \tag{2}$$

where the conditional expectation is taken using the *ad hoc* risk neutral measure. Using iterated conditioning, we can show that the discounted bond price $\mathbb{D}_n \mathbb{B}_{n,M}$ is a martingale, and hence they also satisfy the simpler relations

$$\mathbb{D}_n \mathbb{B}_{n,M} = \widetilde{\mathbb{E}}[\mathbb{D}_{n+1} \mathbb{B}_{n+1,M} \mid \mathcal{F}_n] \ . \tag{3}$$

Using the formula for the discount process, we can simplify the above formula to get

$$\mathbb{B}_{n,M} = \frac{1}{1 + \mathbb{R}_n} \widetilde{\mathbb{E}}[\mathbb{B}_{n+1,M} \mid \mathcal{F}_n] \ . \tag{4}$$

Once we determine the bond prices using the above *ad hoc* risk neutral measure, we can calibrate the model by matching the bond prices at time 0 with data from market observations. The calibration may involve tuning the risk neutral probability measure as well as the parameters of the binomial tree model of short rates. Once the short rate model is calibrated, it can be used for pricing derivative securities based on interest rates. (Of course, if the financial market also consists of other primary assets like stocks, the risk neutral probability measure can no longer be chosen in an *ad hoc* manner but must be chosen such that the discounted stock price processes are martingales with respect to this measure.)

**Note:** You have to show the intermediate steps of the calculations for all the exercises below, and not just write the final answers.

**Exercise 1a (6 points):** Take $N = 3$ and consider the probability measure $\widetilde{\mathbb{P}}$ for the outcomes in the sample space of sequences of $N = 3$ coin tosses:

| $\underline{\omega}$ | $HHH$ | $HHT$ | $HTH$ | $HTT$ | $THH$ | $THT$ | $TTH$ | $TTT$ |
|---|---|---|---|---|---|---|---|---|
| $\widetilde{\mathbb{P}}\{\underline{\omega}\}$ | 2/9 | 1/9 | 1/12 | 1/12 | 1/6 | 1/12 | 1/8 | 1/8 |

In general, the probabilities for heads and tails starting from a particular node is dependent on the particular sequence of coin tosses used to reach that node. Construct the binomial tree for this probability measure by calculating the transition probabilities from one node of the tree to the next. You need to compute all conditional probabilities of form $\widetilde{\mathbb{P}}\{\omega_i \mid \omega_{i-1}, \omega_{i-2}, \ldots, \omega_1\}$ for all $i = 1, \ldots, N$.

**Ans.** First, the probabilities are legitimate: $\frac{2}{9} + \frac{1}{9} + \frac{1}{12} + \frac{1}{12} + \frac{1}{6} + \frac{1}{12} + \frac{1}{8} + \frac{1}{8} = 1$.

To compute the transition probabilities, we need the probabilities of the various events involved in the conditional probability. These are all events fixed by specifying the outcomes of the first $n$ coin tosses. We compute these first. In the binomial tree, these probabilities can be placed at the nodes, and the transition probabilities are placed on the edges connected successive nodes.

For instance, if we consider the event that the first coin toss results in H, then, the subset of sample space is

$$A_H = \{HHH, HHT, HTH, HTT\} \ . \tag{7}$$

The probability of this event is then the sum of the probabilities of the individual outcomes in $A_H$. Graphically, this corresponds to summing over the probabilities of all complete paths that pass through the node that results from a $H$ in the first toss.

In general, the probability at a node reached by a partial path $\omega_1 \cdots \omega_n$ is the sum of the probabilities given in the question for those complete paths which pass through the node.

Thus, the probability of the first toss being $H$ is

$$\mathbb{P}\{A_H\} = \tfrac{2}{9} + \tfrac{1}{9} + \tfrac{1}{12} + \tfrac{1}{12} = \tfrac{1}{3} + \tfrac{1}{6} = \tfrac{1}{2} \ , \tag{8}$$

and similarly, the probability that the first coin toss results in $T$ is

$$\widetilde{\mathbb{P}}\{A_T\} = \tfrac{1}{6} + \tfrac{1}{12} + \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2} \ . \tag{9}$$

We also alternatively denote the events $A_H$ and $A_T$ by the conditions $\omega_1 = H$ and $\omega_2 = T$ respectively. Then, we have

*Time 1.*

$$\begin{aligned} \widetilde{\mathbb{P}}\{\omega_1 = H\} &= \tfrac{2}{9} + \tfrac{1}{9} + \tfrac{1}{12} + \tfrac{1}{12} = \tfrac{1}{3} + \tfrac{1}{6} = \tfrac{1}{2} \ , \\ \widetilde{\mathbb{P}}\{\omega_1 = T\} &= \tfrac{1}{6} + \tfrac{1}{12} + \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2} \ . \end{aligned} \tag{10}$$

The probabilities at the nodes at time 2 can be similarly computed by adding up the complete paths which pass through a node at time 2.

*Time 2.* Node probabilities

$$\begin{aligned} \widetilde{\mathbb{P}}\{\omega_1\omega_2 = HH\} &= \tfrac{2}{9} + \tfrac{1}{9} = \tfrac{1}{3} \ , & \widetilde{\mathbb{P}}\{\omega_1\omega_2 = HT\} &= \tfrac{1}{12} + \tfrac{1}{12} = \tfrac{1}{6} \ , \\ \widetilde{\mathbb{P}}\{\omega_1\omega_2 = TH\} &= \tfrac{1}{6} + \tfrac{1}{12} = \tfrac{1}{4} \ , & \widetilde{\mathbb{P}}\{\omega_1\omega_2 = TT\} &= \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4} \ , \end{aligned} \tag{11}$$

The probabilities at time 3 are the ones given in the question since the process ends at time $N = 3$.

Now, we can compute the the transition probabilities for the second toss given the first toss:

$$\widetilde{\mathbb{P}}\{\omega_2 = H \mid \omega_1 = H\} = \frac{1/3}{1/2} = \frac{2}{3}\,, \qquad \widetilde{\mathbb{P}}\{\omega_2 = T \mid \omega_1 = H\} = \frac{1/6}{1/2} = \frac{1}{3}\,,$$
$$\widetilde{\mathbb{P}}\{\omega_2 = H \mid \omega_1 = T\} = \frac{1/4}{1/2} = \frac{1}{2}\,, \qquad \widetilde{\mathbb{P}}\{\omega_2 = T \mid \omega_1 = T\} = \frac{1/4}{1/2} = \frac{1}{2}\,. \tag{12}$$

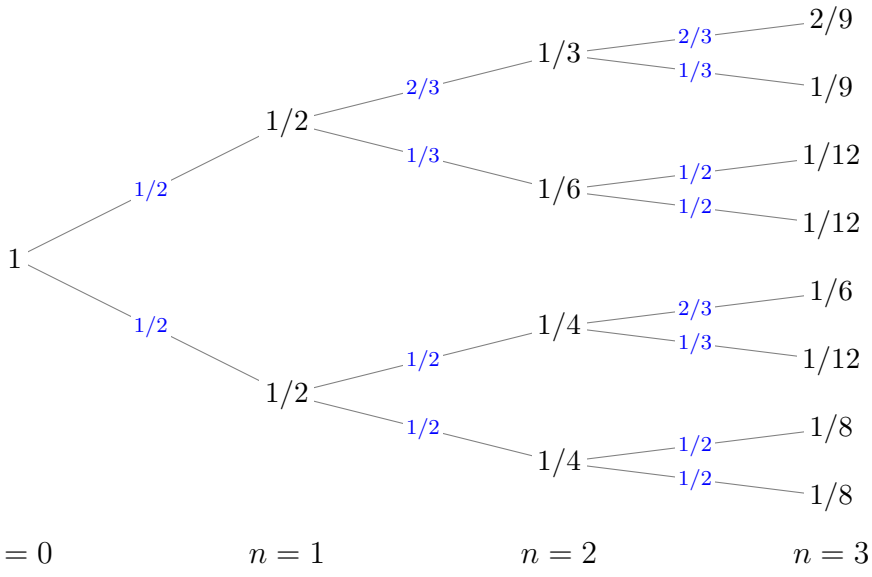
and the transition probabilities for the third toss given the first two tosses:

*Time 3.*

$$\widetilde{\mathbb{P}}\{\omega_3 = H \mid \omega_1\omega_2 = HH\} = \frac{2/9}{1/3} = \frac{2}{3}\,, \qquad \widetilde{\mathbb{P}}\{\omega_3 = T \mid \omega_1\omega_2 = HH\} = \frac{1/9}{1/3} = \frac{1}{3}\,,$$
$$\widetilde{\mathbb{P}}\{\omega_3 = H \mid \omega_1\omega_2 = HT\} = \frac{1/12}{1/6} = \frac{1}{2}\,, \qquad \widetilde{\mathbb{P}}\{\omega_3 = T \mid \omega_1\omega_2 = HT\} = \frac{1/12}{1/6} = \frac{1}{2}\,,$$
$$\widetilde{\mathbb{P}}\{\omega_3 = H \mid \omega_1\omega_2 = TH\} = \frac{1/6}{1/4} = \frac{2}{3}\,, \qquad \widetilde{\mathbb{P}}\{\omega_3 = T \mid \omega_1\omega_2 = TH\} = \frac{1/12}{1/4} = \frac{1}{3}\,,$$
$$\widetilde{\mathbb{P}}\{\omega_3 = H \mid \omega_1\omega_2 = TT\} = \frac{1/8}{1/4} = \frac{1}{2}\,, \qquad \widetilde{\mathbb{P}}\{\omega_3 = T \mid \omega_1\omega_2 = TT\} = \frac{1/8}{1/4} = \frac{1}{2}\,. \tag{13}$$

The pair of probabilities for two outcomes of the third toss given the first two tosses sum to 1, as they must.

The tree below shows the node probabilities at the nodes and the transition probabilities (blue) on the branches:




Unlike the stock models of the earlier assignments, the probabilities are not constant over time: after getting a $H$, the next toss is $H$ with probability 2/3, whereas after getting a $T$, the probability of the next toss being $H$ is 1/2. ■

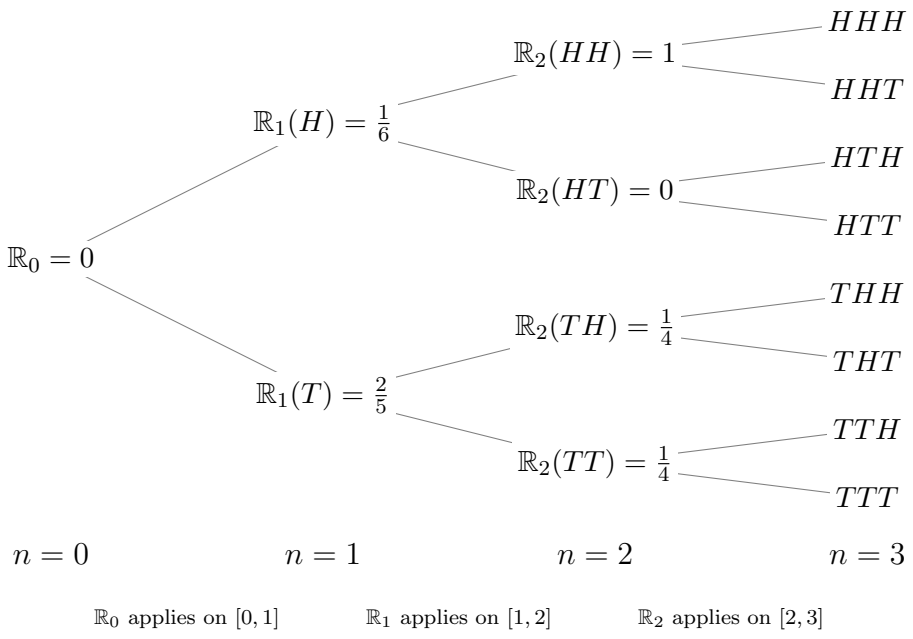
**Exercise 1b (2 points):** Consider the following interest rate process $\mathbb{R}_n$ that holds for the range $0 \leq n \leq 2$:

$$\begin{aligned} &\mathbb{R}_0 = 0 \ , \quad \mathbb{R}_1(H) = 1/6 \ , \quad \mathbb{R}_1(T) = 2/5 \ , \\ &\mathbb{R}_2(HH) = 1 \ , \quad \mathbb{R}_2(HT) = 0 \ , \quad \mathbb{R}_2(TH) = 1/4 \ , \quad \mathbb{R}_2(TT) = 1/4 \ . \end{aligned} \tag{5}$$

The rate $\mathbb{R}_0$ applies in the interval 0 to 1 for both possibilities of the first coin toss. The rate $\mathbb{R}_1$ applies in the interval 1 to 2, with $\mathbb{R}_1(H)$ applying for the two outcomes of $\omega_2$ that follow $\omega_1 = H$, and $\mathbb{R}_1(T)$ applying for the two outcomes of $\omega_2$ that follow $\omega_1 = T$. Finally, $\mathbb{R}_2$ applies in the interval 2 to 3, with the four possible values of $\mathbb{R}_2$ applying for the four different ways in which one can reach time 2 from time 0.

Draw the binomial tree with the above interest rates displayed at the appropriate nodes, with a clear indication as to which time interval they apply to. ■

**Ans.** $\mathbb{R}_n(\omega_1 \cdots \omega_n)$ is known at the node $\omega_1 \cdots \omega_n$ at time $n$, and applies on *both* branches leaving that node, i.e., over $[n, n+1]$. The rates are therefore placed at the nodes $n = 0, 1, 2$; the time-3 nodes carry no rate (none is needed for $N = 3$).




For example, $\mathbb{R}_1(H) = 1/6$ is the one-period rate between times 1 and 2 on both paths $HH\omega_3$ and $HT\omega_3$, while $\mathbb{R}_2(HH) = 1$ is the rate between times 2 and 3 on both paths $HHH$ and $HHT$. ■

**Exercise 1c (8 points):** Fill up the entries in the following table of discounting factors vs. coin toss outcomes. Note that we have ignored the third coin toss since it is not relevant for discounting factors that apply for the periods from time 0 up to time 3.

| $\omega_1\omega_2$ | $\frac{1}{1+\mathbb{R}_0}$ | $\frac{1}{1+\mathbb{R}_1}$ | $\frac{1}{1+\mathbb{R}_2}$ | $\mathbb{D}_1$ | $\mathbb{D}_2$ | $\mathbb{D}_3$ |
|---|---|---|---|---|---|---|
| $HH$ | | | | | | |
| $HT$ | | | | | | |
| $TH$ | | | | | | |
| $TT$ | | | | | | |

**Ans.** The one-period factors are

$$\frac{1}{1+\mathbb{R}_0} = 1 \ , \quad \frac{1}{1+\mathbb{R}_1(H)} = \frac{1}{7/6} = \frac{6}{7} \ , \quad \frac{1}{1+\mathbb{R}_1(T)} = \frac{1}{7/5} = \frac{5}{7} \ , \tag{14}$$

$$\frac{1}{1+\mathbb{R}_2(HH)} = \frac{1}{2} \ , \quad \frac{1}{1+\mathbb{R}_2(HT)} = 1 \ , \quad \frac{1}{1+\mathbb{R}_2(TH)} = \frac{1}{1+\mathbb{R}_2(TT)} = \frac{1}{5/4} = \frac{4}{5} \ . \tag{15}$$

The discount process is built up multiplicatively, $\mathbb{D}_{n+1} = \mathbb{D}_n \frac{1}{1+\mathbb{R}_n}$, so that $\mathbb{D}_n$ depends only on the first $n-1$ tosses ($\mathbb{D}_1$ on none, $\mathbb{D}_2$ on $\omega_1$, $\mathbb{D}_3$ on $\omega_1\omega_2$):

$$\begin{aligned} \mathbb{D}_1 = \tfrac{1}{1+\mathbb{R}_0} = 1 \ , \qquad \mathbb{D}_2(H\cdot) = 1 \cdot \tfrac{6}{7} = \tfrac{6}{7} \ , \qquad \mathbb{D}_2(T\cdot) = 1 \cdot \tfrac{5}{7} = \tfrac{5}{7} \ , \\ \mathbb{D}_3(HH) = \tfrac{6}{7} \cdot \tfrac{1}{2} = \tfrac{3}{7} \ , \quad \mathbb{D}_3(HT) = \tfrac{6}{7} \cdot 1 = \tfrac{6}{7} \ , \\ \mathbb{D}_3(TH) = \tfrac{5}{7} \cdot \tfrac{4}{5} = \tfrac{4}{7} \ , \quad \mathbb{D}_3(TT) = \tfrac{5}{7} \cdot \tfrac{4}{5} = \tfrac{4}{7} \ . \end{aligned} \tag{16}$$

| $\omega_1\omega_2$ | $\frac{1}{1+\mathbb{R}_0}$ | $\frac{1}{1+\mathbb{R}_1}$ | $\frac{1}{1+\mathbb{R}_2}$ | $\mathbb{D}_1$ | $\mathbb{D}_2$ | $\mathbb{D}_3$ |
|---|---|---|---|---|---|---|
| $HH$ | 1 | 6/7 | 1/2 | 1 | 6/7 | 3/7 |
| $HT$ | 1 | 6/7 | 1 | 1 | 6/7 | 6/7 |
| $TH$ | 1 | 5/7 | 4/5 | 1 | 5/7 | 4/7 |
| $TT$ | 1 | 5/7 | 4/5 | 1 | 5/7 | 4/7 |

■

**Exercise 1d (8 points):** Using the risk-neutral transition probabilities from Exercise 1a, and the discounting factors from Exercise 1c, compute the spectrum of bond prices for the zero coupon bonds $\text{ZCB}_1$, $\text{ZCB}_2$, $\text{ZCB}_3$. That is, compute the risk-neutral bond prices $\mathbb{B}_{n,M}$ with $n \leq M$ for each value of $M = 1, 2, 3$, using the risk-neutral pricing formula $\mathbb{D}_n\mathbb{B}_{n,M} = \widetilde{\mathbb{E}}[\mathbb{D}_{n+1}\mathbb{B}_{n+1,M} \mid \mathcal{F}_n]$. Draw the prices on a binomial tree, one such tree for each zero coupon bond. ■

**Ans.** We start from $\mathbb{B}_{M,M} = 1$ and roll backwards. At a node $\omega_1 \cdots \omega_n$ the conditional

expectation averages over the next toss with the transition probabilities of Exercise 1a:

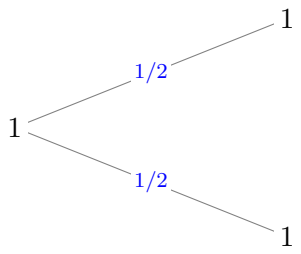
$$\mathbb{D}_n\mathbb{B}_{n,M}(\omega_1\cdots\omega_n) = \sum_{\omega_{n+1}\in\{H,T\}} \widetilde{\mathbb{P}}\{\omega_{n+1} \mid \omega_1\cdots\omega_n\}\,\mathbb{D}_{n+1}(\omega_1\cdots\omega_n)\,\mathbb{B}_{n+1,M}(\omega_1\cdots\omega_{n+1}) \ . \tag{17}$$

Since $\mathbb{D}_{n+1}$ is already known at time $n$, it comes out of the average, and $\mathbb{D}_{n+1}/\mathbb{D}_n = 1/(1+\mathbb{R}_n)$, which is equation (4) of the problem statement. We use the $\mathbb{D}$-form and divide by $\mathbb{D}_n$ at the end.

ZCB$_1$ ($M=1$): We have $\mathbb{B}_{1,1}(H) = \mathbb{B}_{1,1}(T) = 1$. This then gives

$$\mathbb{B}_{0,1} = \tfrac{1}{2}\cdot 1\cdot 1 + \tfrac{1}{2}\cdot 1\cdot 1 = 1 \ . \tag{18}$$

This is just $\mathbb{B}_{0,1} = 1/(1+\mathbb{R}_0)$ with $\mathbb{R}_0 = 0$.




$n=0$ $\qquad\qquad$ $n=1$

ZCB$_2$ ($M=2$): $\mathbb{B}_{2,2} = 1$ at all four time-2 nodes. Then, the prices at time 1 are

*Prices at* $n=1$:

$$\begin{aligned} \mathbb{D}_1\mathbb{B}_{1,2}(H) &= \tfrac{2}{3}\cdot\tfrac{6}{7}\cdot 1 + \tfrac{1}{3}\cdot\tfrac{6}{7}\cdot 1 = \tfrac{6}{7} \quad\Rightarrow\quad \mathbb{B}_{1,2}(H) = \tfrac{6}{7} \approx 0.8571 \ , \\ \mathbb{D}_1\mathbb{B}_{1,2}(T) &= \tfrac{1}{2}\cdot\tfrac{5}{7}\cdot 1 + \tfrac{1}{2}\cdot\tfrac{5}{7}\cdot 1 = \tfrac{5}{7} \quad\Rightarrow\quad \mathbb{B}_{1,2}(T) = \tfrac{5}{7} \approx 0.7143 \ . \end{aligned} \tag{19}$$

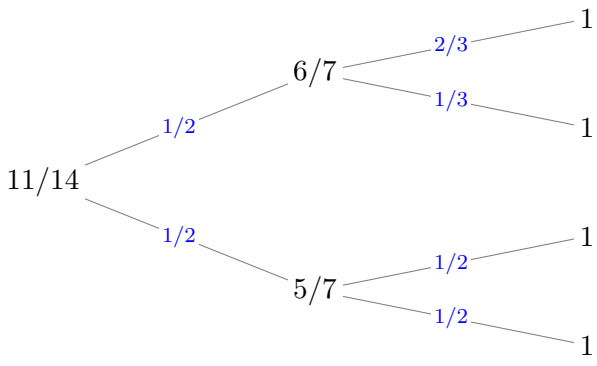
*Price at* $n=0$:

$$\mathbb{B}_{0,2} = \tfrac{1}{2}\cdot 1\cdot\tfrac{6}{7} + \tfrac{1}{2}\cdot 1\cdot\tfrac{5}{7} = \tfrac{11}{14} \approx 0.7857 \ . \tag{20}$$

As a check of our calculations, we can directly calculate from equation (2):

$$\mathbb{B}_{0,2} = \widetilde{\mathbb{E}}[\mathbb{D}_2] = \tfrac{1}{2}\cdot\tfrac{6}{7} + \tfrac{1}{2}\cdot\tfrac{5}{7} = \tfrac{11}{14} \ . \tag{21}$$

The binomial tree is given below:




ZCB$_3$ ($M=3$): We have $\mathbb{B}_{3,3}=1$ at all eight time-3 nodes.

*Prices at* $n=2$: We use the formula $\mathbb{B}_{M-1,M}=1/(1+\mathbb{R}_{M-1})$ that we established in class. Thus,

$$\begin{aligned}
\mathbb{B}_{2,3}(HH)&=\frac{1}{1+\mathbb{R}_2(HH)}=\frac{1}{2}\ , & \mathbb{B}_{2,3}(HT)&=\frac{1}{1+\mathbb{R}_2(HT)}=1\ ,\\
\mathbb{B}_{2,3}(TH)&=\frac{1}{1+\mathbb{R}_2(TH)}=\frac{4}{5}\ , & \mathbb{B}_{2,3}(TT)&=\frac{1}{1+\mathbb{R}_2(TT)}=\frac{4}{5}\ .
\end{aligned}\tag{22}$$

*Prices at* $n=1$: We use the prices $\mathbb{B}_{2,3}$ computed above in the conditional expectation formula:

$$\begin{aligned}
\mathbb{D}_1\mathbb{B}_{1,3}(H)&=\tfrac{2}{3}\,\mathbb{D}_2(H)\,\mathbb{B}_{2,3}(HH)+\tfrac{1}{3}\,\mathbb{D}_2(H)\,\mathbb{B}_{2,3}(HT)=\tfrac{2}{3}\cdot\tfrac{6}{7}\cdot\tfrac{1}{2}+\tfrac{1}{3}\cdot\tfrac{6}{7}\cdot 1=\tfrac{2}{7}+\tfrac{2}{7}=\tfrac{4}{7}\ ,\\
\mathbb{D}_1\mathbb{B}_{1,3}(T)&=\tfrac{1}{2}\,\mathbb{D}_2(T)\,\mathbb{B}_{2,3}(TH)+\tfrac{1}{2}\,\mathbb{D}_2(T)\,\mathbb{B}_{2,3}(TT)=\tfrac{1}{2}\cdot\tfrac{5}{7}\cdot\tfrac{4}{5}+\tfrac{1}{2}\cdot\tfrac{5}{7}\cdot\tfrac{4}{5}=\tfrac{4}{7}\ ,
\end{aligned}\tag{23}$$

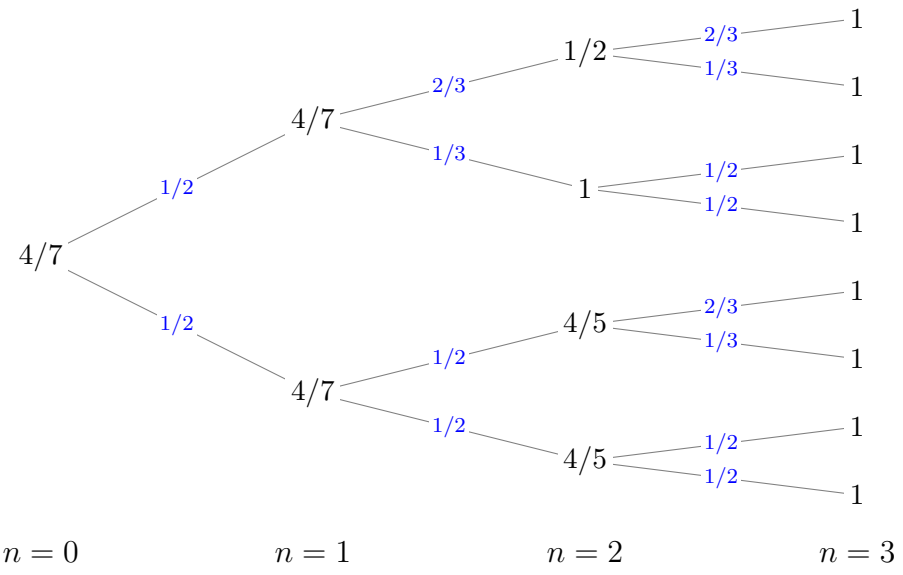
so $\mathbb{B}_{1,3}(H)=\mathbb{B}_{1,3}(T)=4/7\approx 0.5714$.

*Price at* $n=0$: Finally, the price at time 0:

$$\mathbb{B}_{0,3}=\tfrac{1}{2}\cdot 1\cdot\tfrac{4}{7}+\tfrac{1}{2}\cdot 1\cdot\tfrac{4}{7}=\tfrac{4}{7}\approx 0.5714\ .\tag{24}$$

We again check by directly using the definition (2):

$$\begin{aligned}
\mathbb{B}_{0,3}=\widetilde{\mathbb{E}}[\mathbb{D}_3]&=\widetilde{\mathbb{P}}\{HH\}\mathbb{D}_3(HH)+\widetilde{\mathbb{P}}\{HT\}\mathbb{D}_3(HT)+\widetilde{\mathbb{P}}\{TH\}\mathbb{D}_3(TH)+\widetilde{\mathbb{P}}\{TT\}\mathbb{D}_3(TT)\\
&=\tfrac{1}{3}\cdot\tfrac{3}{7}+\tfrac{1}{6}\cdot\tfrac{6}{7}+\tfrac{1}{4}\cdot\tfrac{4}{7}+\tfrac{1}{4}\cdot\tfrac{4}{7}=4\cdot\tfrac{1}{7}=\tfrac{4}{7}\ .
\end{aligned}\tag{25}$$




Note that $\mathbb{B}_{1,3}$ is the same on both time-1 nodes even though the rates differ: at $H$ the current rate is lower (1/6) but the future rate is more likely to be very high ($\mathbb{R}_2(HH) = 1$), while at $T$ the current rate is higher (2/5) but the future rate is moderate (1/4). In this model the two effects exactly cancel. ■

Recall the definition of a forward interest rate $\mathbb{F}_{n,M}$ for $n \leq M$. It is an interest rate that applies between time $M$ and time $M+1$ but is determined at the prior time $n$ itself. It is crucial to note that $\mathbb{F}_{n,M}$ is not a single number but is a random variable that depends on the first $n$ coin tosses. The forward interest rate $\mathbb{F}_{n,M}$ is determined by the bond prices as

$$\mathbb{F}_{n,M} = \frac{\mathbb{B}_{n,M}}{\mathbb{B}_{n,M+1}} - 1 \ . \tag{6}$$

**Exercise 1e (6 points):** Compute the spectrum of the forward rates $\mathbb{F}_{n,M}$ for the range of bond maturities $M = 1, 2$ and the allowed values of $n = 0, 1, \ldots, M$ for a fixed $M$. For this particular model of short interest rates, do you see a trend in the forward rates $\mathbb{F}_{n,M}$ as you vary $n$ for a fixed $M$? How about as you vary $M$ for a fixed $n$? That is, after fixing a time $n$, look at the behaviour of all forward rates $\mathbb{F}_{n,M}$ with $n \leq M \leq N-1$. As always, take $N = 3$.

**Ans.** The set of all forward rates at time $n$ is called the forward rate curve at time $n$.

The forward rate curve at time $n = 0$ consists of

$$\begin{aligned}
\mathbb{F}_{0,0} &= \frac{\mathbb{B}_{0,0}}{\mathbb{B}_{0,1}} - 1 = \frac{1}{1} - 1 = 0 = \mathbb{R}_0 \ , \\
\mathbb{F}_{0,1} &= \frac{\mathbb{B}_{0,1}}{\mathbb{B}_{0,2}} - 1 = \frac{1}{11/14} - 1 = \frac{3}{11} \approx 0.2727 \ , \\
\mathbb{F}_{0,2} &= \frac{\mathbb{B}_{0,2}}{\mathbb{B}_{0,3}} - 1 = \frac{11/14}{4/7} - 1 = \frac{11}{8} - 1 = \frac{3}{8} = 0.375 \ .
\end{aligned} \tag{26}$$

The forward rate curve at time $n = 1$ has two rates $\mathbb{F}_{1,1}$ and $\mathbb{F}_{1,2}$:

$$\mathbb{F}_{1,1}(H) = \frac{1}{6/7} - 1 = \frac{1}{6} \approx 0.1667 \ , \qquad \mathbb{F}_{1,1}(T) = \frac{1}{5/7} - 1 = \frac{2}{5} = 0.4 \ ,$$
$$\mathbb{F}_{1,2}(H) = \frac{6/7}{4/7} - 1 = \frac{1}{2} = 0.5 \ , \qquad \mathbb{F}_{1,2}(T) = \frac{5/7}{4/7} - 1 = \frac{1}{4} = 0.25 \ . \tag{27}$$

The forward rate curve at time $n = 2$ contains only one rate $\mathbb{F}_{2,2} = 1/\mathbb{B}_{2,3} - 1 = \mathbb{R}_2$ with the following values:

$$\mathbb{F}_{2,2}(HH) = 2-1 = 1 \ , \quad \mathbb{F}_{2,2}(HT) = 1-1 = 0 \ , \quad \mathbb{F}_{2,2}(TH) = \mathbb{F}_{2,2}(TT) = \tfrac{5}{4} - 1 = \tfrac{1}{4} \ . \tag{28}$$

Here is a table with the various $M$-forward rates $\mathbb{F}_{n,M}$, with different values of $M$ running horizontally and time $n$ running vertically:

| node | $\mathbb{F}_{n,0}$ $[0,1]$ | $\mathbb{F}_{n,1}$ $[1,2]$ | $\mathbb{F}_{n,2}$ $[2,3]$ |
|---|---|---|---|
| $n = 0$ | 0 | $3/11 \approx 0.273$ | $3/8 = 0.375$ |
| $n = 1$, $H$ | – | $1/6 \approx 0.167$ | $1/2 = 0.5$ |
| $n = 1$, $T$ | – | $2/5 = 0.4$ | $1/4 = 0.25$ |
| $n = 2$, $HH$ / $HT$ | – | – | 1 / 0 |
| $n = 2$, $TH$ / $TT$ | – | – | 1/4 / 1/4 |

Let us look at how the forward rate curve behaves at each time. At $n = 0$, the rates are all numbers and increase with increasing $M$. At $n = 1$, the rates increase when the outcome is $H$ while they decrease when the outcome is $T$. There is no fixed trend which works for both outcomes, and there is too little data to make a general statement. At $n = 2$, there is only one rate and hence no trend can be observed.

Now, for a fixed $M$, if we study the forward rate $M$ for different $n$, we see behaviour that is typical of a binomial model with varying model parameters with time. At each time, the forward rate changes by a random 'shock' depending on the outcome of a coin toss. ■

**Some additional information:** A different starting point of describing interest rates in a financial market is to specify a binomial model for the forward rates and then performing the analysis of the assignment in reverse. The model determines the forward rate curve at each time from which the bond prices are obtained by inverting the formula $\mathbb{F}_{n,M} = \frac{\mathbb{B}_{n,M}}{\mathbb{B}_{n,M+1}} - 1$. The probability measure is now determined by demanding that the bond prices computed from the forward rates form a martingale after discounting.